

Appendix for
Energy Conservation in Tunnel Freezers via Stochastic
Dynamic Programming

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This is an online supplement for "Energy Conservation in Tunnel Freezers via Stochastic Dynamic Programming" where we present complete list of numerical results discussed in Sections 5 and 6 of the paper. Tables 1 through 4 provides the results for $N = 3$ case. The first column stands for the number assigned to the data set, which is used as a label for that data set later. The next five columns list the parameter values specific to the data set. The cost values are given in the order of optimal, traditional and alternative together with the percent cost and energy savings obtained via the optimal policy against the traditional and alternative policies.

Tables 5 through 8 present the results for heuristics and optimal solution for the case of $N = 5$. The numbers in the first column correspond to those in the tables for $N = 3$ and represent the data set used in the experiment. The CPU time used (seconds) to generate the proposed policies by the exact value iteration algorithm and the heuristics are displayed in addition to the cost values. Time figures for Heuristic 2, Heuristic 3 are excluded due to space constraints as those are quite similar to the figures of Heuristic 1. Decomposition of total cost into different components is provided as well as the total system-wide cost values. Note that the penalty cost is not provided for the optimal policy and the Decomposition Heuristic as it can be calculated using the given cost values. Similarly, the switching cost for the naive Heuristics can be calculated using the results for total cost and energy cost. Note that for those heuristics, there is no penalty cost listed as under-freezing is not allowed and that type of cost is not incurred.

Finally, Tables 9 to 12 list the results for the case of $N = 15$, which is studied in Section 6.2 of the paper.

	p	r	R	q	Q	optimal cost	traditional cost	% savings (w.r. to trad.)	% energy sav. (w.r. to trad.)	alternative cost	% savings (w.r. to alt.)	% energy s (w.r. to alt.)
1	1	2	2	0.5	1	250.69	302.5	17.13	26.79	276.87	9.45	14.06
2	1	2	2	0.5	2	266.69	303.5	12.13	26.10	290.06	8.06	13.25
3	1	2	2	0.5	4	295.24	305.5	3.36	19.69	316.45	6.70	5.72
4	1	2	2	0.5	8	303.30	309.5	2.00	2.01	369.22	17.85	-15.02
5	1	2	2	1	1	263.96	304	13.17	25.81	284.98	7.38	12.91
6	1	2	2	1	2	279.36	305	8.41	25.29	298.17	6.31	12.31
7	1	2	2	1	4	300.77	307	2.03	2.26	324.56	7.33	-14.74
8	1	2	2	1	8	304.77	311	2.00	2.01	377.33	19.23	-15.02
9	1	2	2	2	1	283.75	307	7.57	19.75	301.19	5.79	5.80
10	1	2	2	2	2	296.80	308	3.64	20.45	314.39	5.59	6.63
11	1	2	2	2	4	303.71	310	2.03	2.26	340.77	10.88	-14.74
12	1	2	2	2	8	307.71	314	2.00	2.01	393.55	21.81	-15.02
13	1	2	2	4	1	305.82	313	2.29	2.83	333.62	8.33	-14.06
14	1	2	2	4	2	307.28	314	2.14	2.49	346.82	11.40	-14.46
15	1	2	2	4	4	309.57	316	2.03	2.26	373.20	17.05	-14.74
16	1	2	2	4	8	313.59	320	2.00	2.01	425.98	26.38	-15.02
17	1	2	4	0.5	1	250.69	302.5	17.13	26.79	276.87	9.45	14.06
18	1	2	4	0.5	2	266.69	303.5	12.13	26.10	290.06	8.06	13.25
19	1	2	4	0.5	4	295.33	305.5	3.33	19.56	316.45	6.67	5.57
20	1	2	4	0.5	8	303.31	309.5	2.00	2.00	369.22	17.85	-15.03
21	1	2	4	1	1	263.96	304	13.17	25.81	284.98	7.38	12.91
22	1	2	4	1	2	279.36	305	8.41	25.29	298.17	6.31	12.31
23	1	2	4	1	4	300.78	307	2.02	2.22	324.56	7.32	-14.78
24	1	2	4	1	8	304.78	311	2.00	2.00	377.33	19.23	-15.03
25	1	2	4	2	1	283.75	307	7.57	19.75	301.19	5.79	5.80
26	1	2	4	2	2	296.99	308	3.57	19.71	314.39	5.53	5.76
27	1	2	4	2	4	303.72	310	2.03	2.22	340.77	10.87	-14.78
28	1	2	4	2	8	307.72	314	2.00	2.00	393.55	21.81	-15.03
29	1	2	4	4	1	305.88	313	2.28	2.59	333.62	8.32	-14.34
30	1	2	4	4	2	307.31	314	2.13	2.33	346.82	11.39	-14.65
31	1	2	4	4	4	309.59	316	2.03	2.22	373.20	17.05	-14.78
32	1	2	4	4	8	313.60	320	2.00	2.00	425.98	26.38	-15.03
33	1	2	8	0.5	1	250.69	302.5	17.13	26.79	276.87	9.45	14.06
34	1	2	8	0.5	2	266.69	303.5	12.13	26.10	290.06	8.06	13.25
35	1	2	8	0.5	4	295.33	305.5	3.33	19.56	316.45	6.67	5.57
36	1	2	8	0.5	8	303.31	309.5	2.00	2.00	369.22	17.85	-15.04
37	1	2	8	1	1	263.96	304	13.17	25.81	284.98	7.38	12.91
38	1	2	8	1	2	279.36	305	8.41	25.29	298.17	6.31	12.31
39	1	2	8	1	4	300.78	307	2.02	2.22	324.56	7.32	-14.78
40	1	2	8	1	8	304.78	311	2.00	2.00	377.33	19.23	-15.04
41	1	2	8	2	1	283.75	307	7.57	19.75	301.19	5.79	5.80
42	1	2	8	2	2	296.99	308	3.57	19.71	314.39	5.53	5.76
43	1	2	8	2	4	303.72	310	2.03	2.22	340.77	10.87	-14.78
44	1	2	8	2	8	307.72	314	2.00	2.00	393.55	21.81	-15.04
45	1	2	8	4	1	305.88	313	2.28	2.59	333.62	8.32	-14.34
46	1	2	8	4	2	307.31	314	2.13	2.33	346.82	11.39	-14.65
47	1	2	8	4	4	309.59	316	2.03	2.22	373.20	17.05	-14.78
48	1	2	8	4	8	313.60	320	2.00	2.00	425.98	26.38	-15.04
49	1	4	2	0.5	1	250.69	302.5	17.13	26.79	276.87	9.45	14.06
50	1	4	2	0.5	2	266.69	303.5	12.13	26.10	290.06	8.06	13.25
51	1	4	2	0.5	4	295.33	305.5	3.33	19.56	316.45	6.67	5.57
52	1	4	2	0.5	8	303.31	309.5	2.00	2.00	369.22	17.85	-15.03
53	1	4	2	1	1	263.96	304	13.17	25.81	284.98	7.38	12.91
54	1	4	2	1	2	279.36	305	8.41	25.29	298.17	6.31	12.31
55	1	4	2	1	4	300.78	307	2.02	2.22	324.56	7.32	-14.78
56	1	4	2	1	8	304.78	311	2.00	2.00	377.33	19.23	-15.03
57	1	4	2	2	1	283.75	307	7.57	19.75	301.19	5.79	5.80
58	1	4	2	2	2	296.99	308	3.57	19.71	314.39	5.53	5.76
59	1	4	2	2	4	303.72	310	2.03	2.22	340.77	10.87	-14.78
60	1	4	2	2	8	307.72	314	2.00	2.00	393.55	21.81	-15.03
61	1	4	2	4	1	305.88	313	2.28	2.59	333.62	8.32	-14.34
62	1	4	2	4	2	307.31	314	2.13	2.33	346.82	11.39	-14.65
63	1	4	2	4	4	309.59	316	2.03	2.22	373.20	17.05	-14.78
64	1	4	2	4	8	313.60	320	2.00	2.00	425.98	26.38	-15.03
65	1	4	4	0.5	1	250.69	302.5	17.13	26.79	276.87	9.45	14.06
66	1	4	4	0.5	2	266.69	303.5	12.13	26.10	290.06	8.06	13.25
67	1	4	4	0.5	4	295.33	305.5	3.33	19.56	316.45	6.67	5.57
68	1	4	4	0.5	8	303.31	309.5	2.00	2.00	369.22	17.85	-15.04
69	1	4	4	1	1	263.96	304	13.17	25.81	284.98	7.38	12.91
70	1	4	4	1	2	279.36	305	8.41	25.29	298.17	6.31	12.31
71	1	4	4	1	4	300.78	307	2.02	2.22	324.56	7.32	-14.78
72	1	4	4	1	8	304.78	311	2.00	2.00	377.33	19.23	-15.04

Table 1: Results for $N = 3$

	p	r	R	q	Q	optimal cost	traditional cost	% savings (w.r. to trad.)	% energy sav. (w.r. to trad.)	alternative cost	% savings (w.r. to alt.)	% energy s (w.r. to alt.)
73	1	4	4	2	1	283.75	307	7.57	19.75	301.19	5.79	5.80
74	1	4	4	2	2	296.99	308	3.57	19.71	314.39	5.53	5.76
75	1	4	4	2	4	303.72	310	2.03	2.22	340.77	10.87	-14.78
76	1	4	4	2	8	307.72	314	2.00	2.00	393.55	21.81	-15.04
77	1	4	4	4	1	305.88	313	2.28	2.59	333.62	8.32	-14.34
78	1	4	4	4	2	307.31	314	2.13	2.33	346.82	11.39	-14.65
79	1	4	4	4	4	309.59	316	2.03	2.22	373.20	17.05	-14.78
80	1	4	4	4	8	313.60	320	2.00	2.00	425.98	26.38	-15.04
81	1	4	8	0.5	1	250.69	302.5	17.13	26.79	276.87	9.45	14.06
82	1	4	8	0.5	2	266.69	303.5	12.13	26.10	290.06	8.06	13.25
83	1	4	8	0.5	4	295.33	305.5	3.33	19.56	316.45	6.67	5.57
84	1	4	8	0.5	8	303.31	309.5	2.00	2.00	369.22	17.85	-15.04
85	1	4	8	1	1	263.96	304	13.17	25.81	284.98	7.38	12.91
86	1	4	8	1	2	279.36	305	8.41	25.29	298.17	6.31	12.31
87	1	4	8	1	4	300.78	307	2.02	2.22	324.56	7.32	-14.78
88	1	4	8	1	8	304.78	311	2.00	2.00	377.33	19.23	-15.04
89	1	4	8	2	1	283.75	307	7.57	19.75	301.19	5.79	5.80
90	1	4	8	2	2	296.99	308	3.57	19.71	314.39	5.53	5.76
91	1	4	8	2	4	303.72	310	2.03	2.22	340.77	10.87	-14.78
92	1	4	8	2	8	307.72	314	2.00	2.00	393.55	21.81	-15.04
93	1	4	8	4	1	305.88	313	2.28	2.59	333.62	8.32	-14.34
94	1	4	8	4	2	307.31	314	2.13	2.33	346.82	11.39	-14.65
95	1	4	8	4	4	309.59	316	2.03	2.22	373.20	17.05	-14.78
96	1	4	8	4	8	313.60	320	2.00	2.00	425.98	26.38	-15.04
97	2	2	2	0.5	1	467.07	602.5	22.48	28.76	532.45	12.28	16.38
98	2	2	2	0.5	2	485.77	603.5	19.51	28.26	545.64	10.97	15.79
99	2	2	2	0.5	4	515.69	605.5	14.83	29.21	572.03	9.85	16.91
100	2	2	2	0.5	8	558.30	609.5	8.40	33.31	624.80	10.64	21.72
101	2	2	2	1	1	483.43	604	19.96	28.12	540.55	10.57	15.62
102	2	2	2	1	2	500.77	605	17.23	28.47	553.75	9.57	16.03
103	2	2	2	1	4	527.11	607	13.16	31.89	580.13	9.14	20.05
104	2	2	2	1	8	564.48	611	7.61	30.82	632.91	10.81	18.80
105	2	2	2	2	1	510.09	607	15.97	28.09	556.77	8.38	15.59
106	2	2	2	2	2	523.94	608	13.83	30.02	569.96	8.07	17.85
107	2	2	2	2	4	546.26	610	10.45	32.42	596.35	8.40	20.68
108	2	2	2	2	8	570.80	614	7.04	30.06	649.12	12.07	17.90
109	2	2	2	4	1	544.97	613	11.10	27.49	589.20	7.51	14.89
110	2	2	2	4	2	552.12	614	10.08	27.35	602.39	8.35	14.72
111	2	2	2	4	4	564.85	616	8.30	29.21	628.78	10.17	16.90
112	2	2	2	4	8	579.72	620	6.50	32.23	681.55	14.94	20.45
113	2	2	4	0.5	1	467.07	602.5	22.48	28.76	532.45	12.28	16.38
114	2	2	4	0.5	2	485.92	603.5	19.48	27.66	545.64	10.94	15.09
115	2	2	4	0.5	4	519.60	605.5	14.19	26.90	572.03	9.16	14.19
116	2	2	4	0.5	8	575.41	609.5	5.59	26.22	624.80	7.90	13.40
117	2	2	4	1	1	483.47	604	19.96	28.00	540.55	10.56	15.48
118	2	2	4	1	2	501.39	605	17.12	26.78	553.75	9.45	14.06
119	2	2	4	1	4	533.37	607	12.13	26.17	580.13	8.06	13.34
120	2	2	4	1	8	586.57	611	4.00	29.90	632.91	7.32	17.71
121	2	2	4	2	1	510.82	607	15.85	26.02	556.77	8.25	13.16
122	2	2	4	2	2	527.93	608	13.17	25.81	569.96	7.38	12.92
123	2	2	4	2	4	557.61	610	8.59	26.23	596.35	6.50	13.40
124	2	2	4	2	8	598.90	614	2.46	27.06	649.12	7.74	14.39
125	2	2	4	4	1	554.04	613	9.62	19.77	589.20	5.97	5.82
126	2	2	4	4	2	566.09	614	7.80	20.58	602.39	6.03	6.78
127	2	2	4	4	4	586.88	616	4.73	22.77	628.78	6.66	9.34
128	2	2	4	4	8	607.04	620	2.09	2.98	681.55	10.93	-13.88
129	2	2	8	0.5	1	467.07	602.5	22.48	28.76	532.45	12.28	16.38
130	2	2	8	0.5	2	485.92	603.5	19.48	27.66	545.64	10.94	15.09
131	2	2	8	0.5	4	519.60	605.5	14.19	26.89	572.03	9.16	14.18
132	2	2	8	0.5	8	578.64	609.5	5.06	19.56	624.80	7.39	5.57
133	2	2	8	1	1	483.47	604	19.96	28.00	540.55	10.56	15.48
134	2	2	8	1	2	501.39	605	17.12	26.78	553.75	9.45	14.06
135	2	2	8	1	4	533.38	607	12.13	26.10	580.13	8.06	13.25
136	2	2	8	1	8	590.68	611	3.33	19.56	632.91	6.67	5.57
137	2	2	8	2	1	510.82	607	15.84	26.02	556.77	8.25	13.16
138	2	2	8	2	2	527.93	608	13.17	25.81	569.96	7.37	12.91
139	2	2	8	2	4	558.74	610	8.40	25.29	596.35	6.31	12.31
140	2	2	8	2	8	601.58	614	2.02	2.22	649.12	7.32	-14.78
141	2	2	8	4	1	554.04	613	9.62	19.76	589.20	5.97	5.82
142	2	2	8	4	2	567.52	614	7.57	19.75	602.39	5.79	5.80
143	2	2	8	4	4	593.99	616	3.57	19.71	628.78	5.53	5.76
144	2	2	8	4	8	607.45	620	2.02	2.22	681.55	10.87	-14.78

Table 2: Results for $N = 3$ (continued)

	p	r	R	q	Q	optimal cost	traditional cost	% savings (w.r. to trad.)	% energy sav. (w.r. to trad.)	alternative cost	% savings (w.r. to alt.)	% energy s (w.r. to alt.)
145	2	4	2	0.5	1	467.07	602.5	22.48	28.76	532.45	12.28	16.38
146	2	4	2	0.5	2	485.92	603.5	19.48	27.66	545.64	10.94	15.09
147	2	4	2	0.5	4	519.60	605.5	14.19	26.90	572.03	9.16	14.19
148	2	4	2	0.5	8	577.35	609.5	5.27	24.72	624.80	7.59	11.63
149	2	4	2	1	1	483.47	604	19.96	28.00	540.55	10.56	15.48
150	2	4	2	1	2	501.39	605	17.12	26.78	553.75	9.45	14.06
151	2	4	2	1	4	533.38	607	12.13	26.13	580.13	8.06	13.29
152	2	4	2	1	8	589.64	611	3.50	21.31	632.91	6.84	7.63
153	2	4	2	2	1	510.82	607	15.84	26.02	556.77	8.25	13.16
154	2	4	2	2	2	527.93	608	13.17	25.81	569.96	7.37	12.91
155	2	4	2	2	4	557.92	610	8.54	25.73	596.35	6.44	12.81
156	2	4	2	2	8	601.46	614	2.04	2.40	649.12	7.34	-14.57
157	2	4	2	4	1	554.04	613	9.62	19.76	589.20	5.97	5.82
158	2	4	2	4	2	566.10	614	7.80	20.54	602.39	6.02	6.73
159	2	4	2	4	4	589.41	616	4.32	20.51	628.78	6.26	6.69
160	2	4	2	4	8	607.31	620	2.05	2.47	681.55	10.89	-14.48
161	2	4	4	0.5	1	467.07	602.5	22.48	28.76	532.45	12.28	16.38
162	2	4	4	0.5	2	485.92	603.5	19.48	27.66	545.64	10.94	15.09
163	2	4	4	0.5	4	519.60	605.5	14.19	26.89	572.03	9.16	14.18
164	2	4	4	0.5	8	578.52	609.5	5.08	20.75	624.80	7.41	6.98
165	2	4	4	1	1	483.47	604	19.96	28.00	540.55	10.56	15.48
166	2	4	4	1	2	501.39	605	17.12	26.78	553.75	9.45	14.06
167	2	4	4	1	4	533.38	607	12.13	26.10	580.13	8.06	13.25
168	2	4	4	1	8	590.49	611	3.36	19.68	632.91	6.70	5.72
169	2	4	4	2	1	510.82	607	15.84	26.02	556.77	8.25	13.16
170	2	4	4	2	2	527.93	608	13.17	25.81	569.96	7.37	12.91
171	2	4	4	2	4	558.74	610	8.40	25.29	596.35	6.31	12.31
172	2	4	4	2	8	601.56	614	2.03	2.25	649.12	7.33	-14.74
173	2	4	4	4	1	554.04	613	9.62	19.76	589.20	5.97	5.82
174	2	4	4	4	2	567.52	614	7.57	19.75	602.39	5.79	5.80
175	2	4	4	4	4	593.61	616	3.63	20.45	628.78	5.59	6.63
176	2	4	4	4	8	607.42	620	2.03	2.25	681.55	10.88	-14.74
177	2	4	8	0.5	1	467.07	602.5	22.48	28.76	532.45	12.28	16.38
178	2	4	8	0.5	2	485.92	603.5	19.48	27.66	545.64	10.94	15.09
179	2	4	8	0.5	4	519.60	605.5	14.19	26.89	572.03	9.16	14.18
180	2	4	8	0.5	8	578.64	609.5	5.06	19.56	624.80	7.39	5.57
181	2	4	8	1	1	483.47	604	19.96	28.00	540.55	10.56	15.48
182	2	4	8	1	2	501.39	605	17.12	26.78	553.75	9.45	14.06
183	2	4	8	1	4	533.38	607	12.13	26.10	580.13	8.06	13.25
184	2	4	8	1	8	590.68	611	3.33	19.56	632.91	6.67	5.57
185	2	4	8	2	1	510.82	607	15.84	26.02	556.77	8.25	13.16
186	2	4	8	2	2	527.93	608	13.17	25.81	569.96	7.37	12.91
187	2	4	8	2	4	558.74	610	8.40	25.29	596.35	6.31	12.31
188	2	4	8	2	8	601.58	614	2.02	2.22	649.12	7.32	-14.78
189	2	4	8	4	1	554.04	613	9.62	19.76	589.20	5.97	5.82
190	2	4	8	4	2	567.52	614	7.57	19.75	602.39	5.79	5.80
191	2	4	8	4	4	593.99	616	3.57	19.71	628.78	5.53	5.76
192	2	4	8	4	8	607.45	620	2.02	2.22	681.55	10.87	-14.78
193	4	2	2	0.5	1	813.52	1202.5	32.35	53.37	1043.60	21.06	45.93
194	4	2	2	0.5	2	834.20	1203.5	30.69	53.93	1056.79	20.33	48.68
195	4	2	2	0.5	4	862.98	1205.5	28.41	56.28	1083.18	20.66	49.21
196	4	2	2	0.5	8	901.28	1209.5	25.48	56.73	1135.95	20.83	45.89
197	4	2	2	1	1	832.62	1204	30.85	53.90	1051.71	20.18	48.15
198	4	2	2	1	2	849.97	1205	29.46	55.83	1064.90	19.81	49.58
199	4	2	2	1	4	875.09	1207	27.50	57.04	1091.29	20.66	49.79
200	4	2	2	1	8	907.75	1211	25.04	57.22	1144.06	19.51	49.59
201	4	2	2	2	1	859.58	1207	28.78	57.05	1067.92	19.24	49.78
202	4	2	2	2	2	873.12	1208	27.72	57.22	1081.12	19.39	49.94
203	4	2	2	2	4	892.76	1210	26.22	57.35	1107.50	20.89	53.43
204	4	2	2	2	8	917.84	1214	24.40	60.33	1160.28	18.90	50.46
205	4	2	2	4	1	892.41	1213	26.43	57.79	1100.35	19.07	49.41
206	4	2	2	4	2	901.20	1214	25.77	56.90	1113.55	19.75	50.49
207	4	2	2	4	4	914.83	1216	24.77	57.82	1139.93	22.37	60.71
208	4	2	2	4	8	925.93	1220	24.10	66.53	1192.71	18.53	39.13
209	4	2	4	0.5	1	850.23	1202.5	29.29	48.14	1043.60	17.29	39.53
210	4	2	4	0.5	2	874.03	1203.5	27.38	48.49	1056.79	16.05	38.85
211	4	2	4	0.5	4	909.37	1205.5	24.56	47.90	1083.18	15.55	35.09
212	4	2	4	0.5	8	959.27	1209.5	20.69	44.70	1135.95	17.00	40.07
213	4	2	4	1	1	872.92	1204	27.50	48.95	1051.71	16.12	39.38
214	4	2	4	1	2	893.27	1205	25.87	48.35	1064.90	15.21	36.68
215	4	2	4	1	4	925.29	1207	23.34	46.06	1091.29	15.27	35.09
216	4	2	4	1	8	969.39	1211	19.95	44.70	1144.06	15.17	38.10

Table 3: Results for $N = 3$ (continued)

	p	r	R	q	Q	optimal cost	traditional cost	% savings (w.r. to trad.)	% energy sav. (w.r. to trad.)	alternative cost	% savings (w.r. to alt.)	% energy s (w.r. to alt.)
217	4	2	4	2	1	905.95	1207	24.94	47.27	1067.92	14.63	36.48
218	4	2	4	2	2	922.98	1208	23.59	45.89	1081.12	14.31	33.84
219	4	2	4	2	4	949.03	1210	21.57	43.64	1107.50	15.17	36.15
220	4	2	4	2	8	984.28	1214	18.92	45.61	1160.28	13.82	34.29
221	4	2	4	4	1	948.24	1213	21.83	44.02	1100.35	13.71	35.09
222	4	2	4	4	2	960.86	1214	20.85	44.70	1113.55	14.01	36.07
223	4	2	4	4	4	980.25	1216	19.39	45.53	1139.93	15.51	37.25
224	4	2	4	4	8	1007.66	1220	17.40	46.54	1192.71	15.30	24.40
225	4	2	8	0.5	1	883.97	1202.5	26.49	35.60	1043.60	13.89	22.96
226	4	2	8	0.5	2	910.05	1203.5	24.38	34.37	1056.79	12.42	22.73
227	4	2	8	0.5	4	948.64	1205.5	21.31	34.17	1083.18	11.23	25.50
228	4	2	8	0.5	8	1008.43	1209.5	16.62	36.53	1135.95	13.71	21.85
229	4	2	8	1	1	907.49	1204	24.63	33.42	1051.71	12.65	21.51
230	4	2	8	1	2	930.18	1205	22.81	33.13	1064.90	11.54	22.97
231	4	2	8	1	4	965.31	1207	20.02	34.37	1091.29	10.67	25.89
232	4	2	8	1	8	1022.00	1211	15.61	36.86	1144.06	11.79	20.82
233	4	2	8	2	1	941.98	1207	21.96	32.55	1067.92	11.05	20.52
234	4	2	8	2	2	961.64	1208	20.39	32.29	1081.12	10.18	20.18
235	4	2	8	2	4	994.73	1210	17.79	32.00	1107.50	10.05	24.70
236	4	2	8	2	8	1043.68	1214	14.03	35.85	1160.28	9.69	20.23
237	4	2	8	4	1	993.70	1213	18.08	32.04	1100.35	9.26	19.95
238	4	2	8	4	2	1010.39	1214	16.77	31.80	1113.55	8.85	24.30
239	4	2	8	4	4	1039.10	1216	14.55	35.51	1139.93	9.46	25.35
240	4	2	8	4	8	1079.85	1220	11.49	36.41	1192.71	15.05	18.71
241	4	4	2	0.5	1	886.56	1202.5	26.27	30.75	1043.60	13.60	19.39
242	4	4	2	0.5	2	913.09	1203.5	24.13	31.32	1056.79	12.30	18.88
243	4	4	2	0.5	4	949.95	1205.5	21.20	30.90	1083.18	11.78	21.19
244	4	4	2	0.5	8	1002.19	1209.5	17.14	32.86	1135.95	13.45	17.38
245	4	4	2	1	1	910.25	1204	24.40	29.61	1051.71	12.41	18.80
246	4	4	2	1	2	932.74	1205	22.59	30.82	1064.90	11.58	19.17
247	4	4	2	1	4	964.90	1207	20.06	31.14	1091.29	11.37	21.11
248	4	4	2	1	8	1014.03	1211	16.27	32.80	1144.06	11.56	17.24
249	4	4	2	2	1	944.47	1207	21.75	29.50	1067.92	10.93	20.02
250	4	4	2	2	2	962.98	1208	20.28	31.86	1081.12	10.58	21.12
251	4	4	2	2	4	990.34	1210	18.15	32.80	1107.50	10.83	22.72
252	4	4	2	2	8	1034.63	1214	14.78	34.17	1160.28	9.74	23.52
253	4	4	2	4	1	993.17	1213	18.12	34.85	1100.35	9.65	23.48
254	4	4	2	4	2	1006.13	1214	17.12	34.81	1113.55	9.76	23.36
255	4	4	2	4	4	1028.62	1216	15.41	34.71	1139.93	10.40	25.09
256	4	4	2	4	8	1068.70	1220	12.40	36.18	1192.71	15.05	18.71
257	4	4	4	0.5	1	886.56	1202.5	26.27	30.75	1043.60	13.53	17.53
258	4	4	4	0.5	2	913.76	1203.5	24.07	29.74	1056.79	11.86	15.66
259	4	4	4	0.5	4	954.72	1205.5	20.80	28.15	1083.18	10.40	17.19
260	4	4	4	0.5	8	1017.87	1209.5	15.84	29.45	1135.95	13.44	17.24
261	4	4	4	1	1	910.33	1204	24.39	29.49	1051.71	12.28	16.38
262	4	4	4	1	2	934.16	1205	22.48	28.76	1064.90	10.97	15.79
263	4	4	4	1	4	971.55	1207	19.51	28.26	1091.29	9.85	16.91
264	4	4	4	1	8	1031.39	1211	14.83	29.21	1144.06	11.45	16.19
265	4	4	4	2	1	945.69	1207	21.65	28.60	1067.92	10.57	15.62
266	4	4	4	2	2	966.88	1208	19.96	28.12	1081.12	9.57	16.03
267	4	4	4	2	4	1001.56	1210	17.23	28.47	1107.50	9.14	20.05
268	4	4	4	2	8	1054.24	1214	13.16	31.89	1160.28	8.92	14.61
269	4	4	4	4	1	1002.20	1213	17.38	27.25	1100.35	8.38	15.59
270	4	4	4	4	2	1020.19	1214	15.96	28.09	1113.55	8.07	17.85
271	4	4	4	4	4	1047.90	1216	13.82	30.01	1139.93	8.40	20.68
272	4	4	4	4	8	1092.52	1220	10.45	32.42	1192.71	15.05	18.71
273	4	4	8	0.5	1	886.56	1202.5	26.27	30.75	1043.60	13.53	17.53
274	4	4	8	0.5	2	913.76	1203.5	24.07	29.74	1056.79	11.85	15.32
275	4	4	8	0.5	4	954.85	1205.5	20.79	27.86	1083.18	9.89	14.31
276	4	4	8	0.5	8	1023.58	1209.5	15.37	27.00	1135.95	13.44	17.24
277	4	4	8	1	1	910.33	1204	24.39	29.49	1051.71	12.28	16.38
278	4	4	8	1	2	934.16	1205	22.48	28.76	1064.90	10.94	15.09
279	4	4	8	1	4	971.85	1207	19.48	27.66	1091.29	9.16	14.19
280	4	4	8	1	8	1039.21	1211	14.19	26.89	1144.06	11.44	16.16
281	4	4	8	2	1	945.71	1207	21.65	28.58	1067.92	10.56	15.48
282	4	4	8	2	2	966.95	1208	19.95	28.00	1081.12	9.45	14.06
283	4	4	8	2	4	1002.80	1210	17.12	26.78	1107.50	8.06	13.34
284	4	4	8	2	8	1066.75	1214	12.13	26.17	1160.28	8.91	14.24
285	4	4	8	4	1	1002.33	1213	17.37	26.94	1100.35	8.25	13.16
286	4	4	8	4	2	1021.65	1214	15.84	26.02	1113.55	7.38	12.92
287	4	4	8	4	4	1055.86	1216	13.17	25.81	1139.93	6.50	13.40
288	4	4	8	4	8	1115.23	1220	8.59	26.22	1192.71	11.25	11.09

Table 4: Results for $N = 3$ (continued)

	optimal				Heuristic 1			Heuristic 2		Heuristic 3		Decomposition Heuristic			
	time (sec)	total cost	energy cost	switch. cost	time (sec)	total cost	energy cost	total cost	energy cost	total cost	energy cost	time (sec)	total cost	energy cost	switch. cost
1	136.78	178.21	161.80	16.41	0.06	192.32	177.63	179.94	168.70	188.98	153.52	0.83	178.52	162.31	16.21
2	137.91	186.53	167.09	19.44	0.06	201.97	177.67	187.11	168.74	210.41	153.61	0.83	191.96	181.77	10.19
3	138.16	198.32	187.78	10.53	0.06	221.27	177.73	201.41	168.79	253.25	153.73	0.83	198.64	193.77	4.87
4	139.17	202.54	193.77	8.76	0.06	259.88	177.85	230.05	168.89	338.94	153.88	0.83	202.54	193.77	8.76
5	135.95	183.63	164.62	19.01	0.05	197.43	177.65	184.09	168.72	203.20	153.58	0.83	191.07	175.16	15.91
6	137.05	191.16	168.23	22.93	0.06	207.08	177.68	191.24	168.75	224.62	153.65	0.83	197.67	193.77	3.90
7	138.36	199.62	193.76	5.85	0.06	226.38	177.75	205.56	168.81	267.47	153.76	0.83	199.62	193.77	5.84
8	138.94	203.51	193.77	9.74	0.05	264.98	177.86	234.19	168.90	353.15	153.90	0.83	203.51	193.77	9.74
9	137.08	192.27	169.02	23.25	0.05	207.65	177.68	192.38	168.75	231.64	153.67	0.83	198.62	193.33	5.29
10	137.97	199.42	168.91	30.08	0.06	217.30	177.72	199.53	168.78	253.06	153.73	0.83	199.62	193.77	5.84
11	139.16	201.56	193.76	7.80	0.05	236.60	177.78	213.84	168.83	295.90	153.81	0.83	201.56	193.77	7.79
12	138.92	205.46	193.77	11.69	0.06	275.21	177.88	242.47	168.92	381.59	153.93	0.83	205.46	193.77	11.69
13	138.17	202.47	193.28	9.18	0.06	228.09	177.75	208.94	168.81	288.50	153.80	0.83	202.50	193.33	9.17
14	138.19	203.50	193.69	9.81	0.05	237.74	177.78	216.10	168.83	309.93	153.83	0.81	203.50	193.76	9.75
15	138.38	205.46	193.76	11.69	0.06	257.04	177.83	230.42	168.89	352.77	153.90	0.83	205.46	193.77	11.69
16	138.94	209.35	193.77	15.58	0.06	295.65	177.92	259.05	168.97	438.47	153.98	0.83	209.35	193.77	15.58
17	135.17	178.21	161.80	16.41	0.05	192.32	177.63	179.94	168.70	188.98	153.52	0.84	178.52	162.31	16.21
18	136.59	186.53	167.09	19.44	0.05	201.97	177.67	187.11	168.74	210.41	153.61	0.84	191.96	181.77	10.19
19	138.41	198.32	187.78	10.53	0.05	221.27	177.73	201.41	168.79	253.25	153.73	0.84	198.64	193.77	4.87
20	138.88	202.54	193.77	8.76	0.06	259.88	177.85	230.05	168.89	338.94	153.88	0.84	202.54	193.77	8.76
21	135.92	183.63	164.62	19.01	0.06	197.43	177.65	184.09	168.72	203.20	153.58	0.84	191.07	175.16	15.91
22	137.05	191.16	168.23	22.93	0.05	207.08	177.68	191.24	168.75	224.62	153.65	0.84	197.67	193.77	3.90
23	138.30	199.62	193.76	5.85	0.06	226.38	177.75	205.56	168.81	267.47	153.76	0.86	199.62	193.77	5.84
24	138.84	203.51	193.77	9.74	0.06	264.98	177.86	234.19	168.90	353.15	153.90	0.84	203.51	193.77	9.74
25	137.05	192.27	169.02	23.25	0.06	207.65	177.68	192.38	168.75	231.64	153.67	0.84	198.62	193.33	5.29
26	137.91	199.42	169.06	30.36	0.06	217.30	177.72	199.53	168.78	253.06	153.73	0.84	199.62	193.77	5.84
27	138.36	201.56	193.76	7.80	0.05	236.60	177.78	213.84	168.83	295.90	153.81	0.84	201.56	193.77	7.79
28	138.88	205.46	193.77	11.69	0.05	275.21	177.88	242.47	168.92	381.59	153.93	0.84	205.46	193.77	11.69
29	138.16	202.47	193.29	9.18	0.06	228.09	177.75	208.94	168.81	288.50	153.80	0.84	202.50	193.33	9.17
30	138.53	203.51	193.69	9.81	0.05	237.74	177.78	216.10	168.83	309.93	153.83	0.84	203.50	193.76	9.75
31	138.34	205.46	193.76	11.69	0.05	257.04	177.83	230.42	168.89	352.77	153.90	0.84	205.46	193.77	11.69
32	138.95	209.35	193.77	15.58	0.06	295.65	177.92	259.05	168.97	438.47	153.98	0.86	209.35	193.77	15.58
33	135.17	178.21	161.80	16.41	0.05	192.32	177.63	179.94	168.70	188.98	153.52	0.89	178.52	162.31	16.21
34	136.52	186.53	167.09	19.44	0.06	201.97	177.67	187.11	168.74	210.41	153.61	0.89	191.96	181.77	10.19
35	138.09	198.32	187.78	10.53	0.06	221.27	177.73	201.41	168.79	253.25	153.73	0.89	198.64	193.77	4.87
36	138.86	202.54	193.77	8.76	0.06	259.88	177.85	230.05	168.89	338.94	153.88	0.88	202.54	193.77	8.76
37	135.97	183.63	164.62	19.01	0.06	197.43	177.65	184.09	168.72	203.20	153.58	0.88	191.07	175.16	15.91
38	137.06	191.16	168.23	22.93	0.06	207.08	177.68	191.24	168.75	224.62	153.65	0.88	197.67	193.77	3.90
39	138.36	199.62	193.76	5.85	0.06	226.38	177.75	205.56	168.81	267.47	153.76	0.88	199.62	193.77	5.84
40	138.86	203.51	193.77	9.74	0.06	264.98	177.86	234.19	168.90	353.15	153.90	0.89	203.51	193.77	9.74
41	137.05	192.27	169.02	23.25	0.06	207.65	177.68	192.38	168.75	231.64	153.67	0.89	198.62	193.33	5.29
42	137.98	199.42	169.06	30.36	0.05	217.30	177.72	199.53	168.78	253.06	153.73	0.89	199.62	193.77	5.84
43	138.34	201.56	193.76	7.80	0.05	236.60	177.78	213.84	168.83	295.90	153.81	0.89	201.56	193.77	7.79
44	138.92	205.46	193.77	11.69	0.06	275.21	177.88	242.47	168.92	381.59	153.93	0.88	205.46	193.77	11.69
45	138.13	202.47	193.29	9.18	0.05	228.09	177.75	208.94	168.81	288.50	153.80	0.88	202.50	193.33	9.17
46	138.22	203.51	193.69	9.81	0.06	237.74	177.78	216.10	168.83	309.93	153.83	0.88	203.50	193.76	9.75
47	138.33	205.46	193.76	11.69	0.06	257.04	177.83	230.42	168.89	352.77	153.90	0.88	205.46	193.77	11.69
48	138.86	209.35	193.77	15.58	0.05	295.65	177.92	259.05	168.97	438.47	153.98	0.88	209.35	193.77	15.58
49	135.17	178.21	161.80	16.41	0.05	192.32	177.63	179.94	168.70	188.98	153.52	0.88	178.52	162.31	16.21
50	136.44	186.53	167.09	19.44	0.05	201.97	177.67	187.11	168.74	210.41	153.61	0.86	191.96	181.77	10.19
51	137.81	198.32	187.78	10.53	0.05	221.27	177.73	201.41	168.79	253.25	153.73	0.86	198.64	193.77	4.87
52	138.33	202.54	193.77	8.76	0.05	259.88	177.85	230.05	168.89	338.94	153.88	0.86	202.54	193.77	8.76
53	135.92	183.63	164.62	19.01	0.06	197.43	177.65	184.09	168.72	203.20	153.58	0.88	191.07	175.16	15.91
54	136.97	191.16	168.23	22.93	0.06	207.08	177.68	191.24	168.75	224.62	153.65	0.88	197.67	193.77	3.90
55	138.08	199.62	193.76	5.85	0.05	226.38	177.75	205.56	168.81	267.47	153.76	0.86	199.62	193.77	5.84
56	138.41	203.51	193.77	9.74	0.05	264.98	177.86	234.19	168.90	353.15	153.90	0.88	203.51	193.77	9.74
57	137.13	192.27	169.02	23.25	0.05	207.65	177.68	192.38	168.75	231.64	153.67	0.88	198.62	193.33	5.29
58	137.80	199.42	169.06	30.36	0.06	217.30	177.72	199.53	168.78	253.06	153.73	0.88	199.62	193.77	5.84
59	138.06	201.56	193.76	7.80	0.06	236.60	177.78	213.84	168.83	295.90	153.81	0.88	201.56	193.77	7.79
60	138.33	205.46	193.77	11.69	0.05	275.21	177.88	242.47	168.92	381.59	153.93	0.88	205.46	193.77	11.69
61	138.05	202.47	193.29	9.18	0.05	228.09	177.75	208.94	168.81	288.50	153.80	0.88	202.50	193.33	9.17
62	138.02	203.51	193.69	9.81	0.06	237.74	177.78	216.10	168.83	309.93	153.83	0.88	203.50	193.76	9.75
63	138.09	205.46	193.76	11.69	0.06	257.04	177.83	230.42	168.89	352.77	153.90	0.88	205.46	193.77	11.69
64	138.33	209.35	193.77	15.58	0.06	295.65	177.92	259.05	168.97	438.47	153.98	0.86	209.35	193.77	15.58
65	135.16	178.21	161.80	16.41	0.06	192.32	177.63	179.94	168.70	188.98	153.52	0.89	178.52	162.31	16.21
66	136.44	186.53	167.09	19.44	0.06	201.97	177.67	187.11	168.74	210.41	153.61	0.89	191.96	181.77	10.19
67	137.80	198.32	187.78	10.53	0.06	221.27	177.73	201.41	168.79	253.25	153.73	0.89	198.64	193.77	4.87
68	138.33	202.54	193.77	8.76	0.05	259.88	177.85	230.05	168.89	338.94	153.88	0.89	202.54	193.77	8.76
69	135.91	183.63	164.62	19.01	0.05	197.43	177.65	184.09	168.72	203.20	153.58	0.88	191.07	175.16	15.91
70	136.97	191.16	168.23	22.93	0.05	207.08	177.68	191.24	168.75	224.62	153.65	0.89	197.67	193.77	3.90
71	138.06	199.62	193.76	5.85	0.06	226.38	177.75	205.56	168.81	267.47	153.76				

	optimal				Heuristic 1			Heuristic 2		Heuristic 3		Decomposition Heuristic			
	time (sec)	total cost	energy cost	switch. cost	time (sec)	total cost	energy cost	total cost	energy cost	total cost	energy cost	time (sec)	total cost	energy cost	switch. cost
73	136.95	192.27	169.02	23.25	0.06	207.65	177.68	192.38	168.75	231.64	153.67	0.89	198.62	193.33	5.29
74	137.74	199.42	169.06	30.36	0.05	217.30	177.72	199.53	168.78	253.06	153.73	0.89	199.62	193.77	5.84
75	138.08	201.56	193.76	7.80	0.06	236.60	177.78	213.84	168.83	295.90	153.81	0.89	201.56	193.77	7.79
76	138.33	205.46	193.77	11.69	0.05	275.21	177.88	242.47	168.92	381.59	153.93	0.89	205.46	193.77	11.69
77	138.03	202.47	193.29	9.18	0.05	228.09	177.75	208.94	168.81	288.50	153.80	0.89	202.50	193.33	9.17
78	137.98	203.51	193.69	9.81	0.05	237.74	177.78	216.10	168.83	309.93	153.83	0.89	203.50	193.76	9.75
79	138.06	205.46	193.76	11.69	0.06	257.04	177.83	230.42	168.89	352.77	153.90	0.89	205.46	193.77	11.69
80	138.34	209.35	193.77	15.58	0.05	295.65	177.92	259.05	168.97	438.47	153.98	0.88	209.35	193.77	15.58
81	135.16	178.21	161.80	16.41	0.05	192.32	177.63	179.94	168.70	188.98	153.52	0.91	178.52	162.31	16.21
82	136.45	186.53	167.09	19.44	0.05	201.97	177.67	187.11	168.74	210.41	153.61	0.92	191.96	181.77	10.19
83	138.05	198.32	187.78	10.53	0.05	221.27	177.73	201.41	168.79	253.25	153.73	0.91	198.64	193.77	4.87
84	138.36	202.54	193.77	8.76	0.06	259.88	177.85	230.05	168.89	338.94	153.88	0.92	202.54	193.77	8.76
85	135.91	183.63	164.62	19.01	0.06	197.43	177.65	184.09	168.72	203.20	153.58	0.91	191.07	175.16	15.91
86	136.95	191.16	168.23	22.93	0.05	207.08	177.68	191.24	168.75	224.62	153.65	0.91	197.67	193.77	3.90
87	138.05	199.62	193.76	5.85	0.05	226.38	177.75	205.56	168.81	267.47	153.76	0.91	199.62	193.77	5.84
88	138.31	203.51	193.77	9.74	0.06	264.98	177.86	234.19	168.90	353.15	153.90	0.91	203.51	193.77	9.74
89	136.98	192.27	169.02	23.25	0.05	207.65	177.68	192.38	168.75	231.64	153.67	0.91	198.62	193.33	5.29
90	137.75	199.42	169.06	30.36	0.06	217.30	177.72	199.53	168.78	253.06	153.73	0.91	199.62	193.77	5.84
91	138.08	201.56	193.76	7.80	0.06	236.60	177.78	213.84	168.83	295.90	153.81	0.91	201.56	193.77	7.79
92	138.39	205.46	193.77	11.69	0.05	275.21	177.88	242.47	168.92	381.59	153.93	0.91	205.46	193.77	11.69
93	138.02	202.47	193.29	9.18	0.05	228.09	177.75	208.94	168.81	288.50	153.80	0.91	202.50	193.33	9.17
94	138.03	203.51	193.69	9.81	0.06	237.74	177.78	216.10	168.83	309.93	153.83	0.91	203.50	193.76	9.75
95	138.06	205.46	193.76	11.69	0.05	257.04	177.83	230.42	168.89	352.77	153.90	0.91	205.46	193.77	11.69
96	138.33	209.35	193.77	15.58	0.06	295.65	177.92	259.05	168.97	438.47	153.98	0.91	209.35	193.77	15.58
97	153.91	338.49	315.04	23.45	0.05	370.87	356.15	349.58	338.31	343.32	307.78	0.88	341.88	311.95	29.93
98	155.61	350.56	320.46	29.84	0.05	380.53	356.17	356.74	338.33	364.75	307.84	0.88	351.62	315.81	32.04
99	155.74	368.57	328.97	35.32	0.06	399.83	356.21	371.06	338.37	407.58	307.92	0.88	377.12	338.25	30.02
100	158.27	394.73	357.31	30.12	0.06	438.43	356.28	399.69	338.43	493.28	308.05	0.88	397.24	385.53	8.70
101	153.61	346.30	319.50	26.79	0.06	375.98	356.16	353.73	338.32	357.54	307.82	0.88	346.89	317.15	29.75
102	154.56	357.34	323.16	32.39	0.06	385.64	356.18	360.89	338.34	378.96	307.86	0.88	358.36	321.39	32.38
103	156.27	373.58	329.34	39.40	0.05	404.94	356.22	375.20	338.38	421.81	307.95	0.89	385.04	357.04	20.52
104	158.31	397.30	370.96	20.80	0.06	443.54	356.29	403.83	338.43	507.50	308.07	0.89	398.21	385.53	9.66
105	155.10	358.97	323.84	34.10	0.05	386.21	356.18	362.01	338.34	385.97	307.88	0.88	359.89	325.69	34.18
106	156.06	367.88	326.42	36.57	0.06	395.86	356.20	369.17	338.36	407.39	307.92	0.88	369.07	323.41	38.28
107	156.92	382.51	331.61	43.68	0.06	415.16	356.24	383.48	338.39	450.24	307.99	0.88	396.30	385.57	7.85
108	158.55	400.05	385.62	11.61	0.05	453.76	356.31	412.11	338.45	535.93	308.10	0.88	400.14	385.53	11.60
109	155.93	377.51	334.67	40.48	0.05	406.64	356.22	378.59	338.38	442.84	307.98	0.88	396.80	387.57	9.17
110	156.55	384.64	333.49	44.28	0.06	416.29	356.24	385.74	338.39	464.27	308.01	0.89	398.18	385.50	9.98
111	158.33	398.09	328.02	55.30	0.05	435.59	356.27	400.06	338.42	507.12	308.07	0.89	400.17	385.57	11.72
112	158.58	403.91	385.62	15.48	0.06	474.20	356.34	428.68	338.47	592.80	308.15	0.89	404.01	385.53	15.46
113	152.99	338.49	315.04	23.45	0.06	370.87	356.15	349.58	338.31	343.32	307.78	0.91	341.88	311.95	29.93
114	154.14	350.57	320.59	29.98	0.06	380.53	356.17	356.74	338.33	364.75	307.84	0.95	351.33	319.16	32.16
115	155.70	368.90	332.47	36.42	0.05	399.83	356.21	371.06	338.37	407.58	307.92	0.89	376.68	346.35	30.34
116	158.22	395.14	366.45	28.56	0.05	438.43	356.28	399.69	338.43	493.28	308.05	0.89	397.30	388.53	8.76
117	153.58	346.30	319.51	26.79	0.05	375.98	356.16	353.73	338.32	357.54	307.82	0.89	346.89	317.15	29.75
118	154.52	357.42	324.51	32.92	0.06	385.64	356.18	360.89	338.34	378.96	307.86	1.05	358.05	325.53	32.51
119	156.10	374.05	334.98	38.87	0.05	404.94	356.22	375.20	338.38	421.81	307.95	0.89	384.85	364.26	20.55
120	158.27	397.62	376.51	21.07	0.06	443.54	356.29	403.83	338.43	507.50	308.07	0.89	398.27	388.53	9.74
121	154.91	359.10	324.27	34.83	0.06	386.21	356.18	362.01	338.34	385.97	307.88	0.89	359.89	325.70	34.19
122	155.69	368.25	330.13	38.11	0.05	395.86	356.20	369.17	338.36	407.39	307.92	0.89	368.92	327.75	41.17
123	156.72	383.24	337.07	45.56	0.05	415.16	356.24	383.48	338.39	450.24	307.99	0.91	396.32	388.51	7.81
124	158.53	404.12	388.50	15.59	0.05	474.20	356.34	428.68	338.47	592.80	308.15	0.89	404.12	388.53	15.58
125	155.83	378.04	338.07	39.32	0.05	406.64	356.22	378.59	338.38	442.84	307.98	0.89	396.83	387.65	9.17
126	156.50	385.45	338.62	46.18	0.06	416.29	356.24	385.74	338.39	464.27	308.01	0.89	398.23	387.65	10.57
127	158.05	399.56	337.74	59.46	0.06	435.59	356.27	400.06	338.42	507.12	308.07	0.89	400.21	388.51	11.70
128	158.53	404.12	388.50	15.59	0.05	474.20	356.34	428.68	338.47	592.80	308.15	0.89	404.12	388.53	15.58
129	152.97	338.49	315.04	23.45	0.06	370.87	356.15	349.58	338.31	343.32	307.78	0.92	341.88	311.95	29.93
130	154.11	350.57	320.59	29.98	0.06	380.53	356.17	356.74	338.33	364.75	307.84	0.91	351.33	319.16	32.16
131	155.70	368.90	332.47	36.42	0.05	399.83	356.21	371.06	338.37	407.58	307.92	0.92	376.68	346.35	30.34
132	158.19	395.16	367.54	27.62	0.06	438.43	356.28	399.69	338.43	493.28	308.05	0.91	397.30	388.54	8.76
133	153.73	346.30	319.51	26.79	0.06	375.98	356.16	353.73	338.32	357.54	307.82	0.92	346.89	317.15	29.75
134	154.54	357.42	324.51	32.92	0.06	385.64	356.18	360.89	338.34	378.96	307.86	0.92	358.05	325.53	32.51
135	156.11	374.05	335.07	38.98	0.05	404.94	356.22	375.20	338.38	421.81	307.95	0.92	384.84	364.27	20.57
136	158.24	397.63	376.53	21.10	0.06	443.54	356.29	403.83	338.43	507.50	308.07	0.92	398.27	388.54	9.74
137	154.76	359.10	324.27	34.83	0.05	386.21	356.18	362.01	338.34	385.97	307.88	0.91	359.89	325.70	34.19
138	155.35	368.25	330.13	38.11	0.06	395.86	356.20	369.17	338.36	407.39	307.92	0.92	368.92	327.75	41.17
139	156.74	383.30	337.34	45.97	0.05	415.16	356.24	383.48	338.39	450.24	307.99	0.92	396.32	388.51	7.81
140	158.46	400.22	388.52	11.70	0.05	453.76	356.31	412.11	338.45	535.93	308.10	0.92	400.22	388.54	11.69
141	155.86	378.05	338.37	39.67	0.05	406.64	356.22	378.59	338.38	442.84	307.98	0.91	396.83	387.65	9.17
142	156.49	385.53	338.93	46.60	0.05	416.29	356.24	385.74	338.39	464.27	308.01	0.92	398.23	387.65	10.57
143	158.02	399.85	338.98												

	optimal				Heuristic 1			Heuristic 2		Heuristic 3		Decomposition Heuristic			
	time (sec)	total cost	energy cost	switch. cost	time (sec)	total cost	energy cost	total cost	energy cost	total cost	energy cost	time (sec)	total cost	energy cost	switch. cost
145	152.75	338.49	315.04	23.45	0.06	370.87	356.15	349.58	338.31	343.32	307.78	0.91	341.88	311.95	29.93
146	153.60	350.57	320.59	29.98	0.06	380.53	356.17	356.74	338.33	364.75	307.84	0.91	351.82	318.75	33.07
147	154.99	368.90	332.47	36.42	0.06	399.83	356.21	371.06	338.37	407.58	307.92	0.91	376.80	346.54	30.10
148	156.67	395.14	366.99	28.08	0.06	438.43	356.28	399.69	338.43	493.28	308.05	0.92	397.30	388.53	8.76
149	153.10	346.30	319.51	26.79	0.06	375.98	356.16	353.73	338.32	357.54	307.82	0.91	346.89	317.15	29.75
150	153.88	357.42	324.51	32.92	0.05	385.64	356.18	360.89	338.34	378.96	307.86	0.91	358.05	325.53	32.51
151	155.22	374.05	334.98	38.87	0.05	404.94	356.22	375.20	338.38	421.81	307.95	0.91	384.93	364.49	20.40
152	156.72	397.62	376.53	21.07	0.06	443.54	356.29	403.83	338.43	507.50	308.07	0.91	398.27	388.53	9.74
153	154.19	359.10	324.27	34.83	0.06	386.21	356.18	362.01	338.34	385.97	307.88	0.91	361.31	323.72	37.59
154	154.66	368.25	330.13	38.11	0.06	395.86	356.20	369.17	338.36	407.39	307.92	0.91	383.13	351.24	31.88
155	155.75	383.24	337.07	45.56	0.05	415.16	356.24	383.48	338.39	450.24	307.99	0.91	396.32	388.53	7.80
156	156.95	400.22	388.52	11.70	0.05	453.76	356.31	412.11	338.45	535.93	308.10	0.91	400.22	388.53	11.69
157	155.20	378.04	338.07	39.32	0.06	406.64	356.22	378.59	338.38	442.84	307.98	0.91	396.83	387.65	9.17
158	155.75	385.45	338.62	46.18	0.06	416.29	356.24	385.74	338.39	464.27	308.01	0.91	398.23	387.65	10.57
159	156.65	399.62	338.63	60.22	0.06	435.59	356.27	400.06	338.42	507.12	308.07	0.91	400.22	388.53	11.69
160	157.01	404.12	388.52	15.59	0.06	474.20	356.34	428.68	338.47	592.80	308.15	0.91	404.12	388.53	15.58
161	152.58	338.49	315.04	23.45	0.06	370.87	356.15	349.58	338.31	343.32	307.78	0.91	341.88	311.95	29.93
162	153.65	350.57	320.59	29.98	0.06	380.53	356.17	356.74	338.33	364.75	307.84	0.91	351.82	318.75	33.07
163	154.98	368.90	332.47	36.42	0.06	399.83	356.21	371.06	338.37	407.58	307.92	0.91	376.76	346.55	30.21
164	156.74	395.15	367.04	28.10	0.06	438.43	356.28	399.69	338.43	493.28	308.05	0.91	397.30	388.54	8.76
165	153.08	346.30	319.51	26.79	0.06	375.98	356.16	353.73	338.32	357.54	307.82	0.91	346.89	317.15	29.75
166	153.86	357.42	324.51	32.92	0.06	385.64	356.18	360.89	338.34	378.96	307.86	0.92	358.05	325.53	32.51
167	155.22	374.05	335.07	38.98	0.05	404.94	356.22	375.20	338.38	421.81	307.95	0.92	384.92	364.50	20.43
168	156.94	397.63	376.53	21.09	0.05	443.54	356.29	403.83	338.43	507.50	308.07	0.92	398.27	388.54	9.74
169	154.14	359.10	324.27	34.83	0.05	386.21	356.18	362.01	338.34	385.97	307.88	0.92	361.31	323.72	37.59
170	154.69	368.25	330.13	38.11	0.05	395.86	356.20	369.17	338.36	407.39	307.92	0.92	383.13	351.24	31.88
171	155.74	383.30	337.34	45.97	0.06	415.16	356.24	383.48	338.39	450.24	307.99	0.92	396.32	388.53	7.80
172	156.94	400.22	388.52	11.70	0.06	453.76	356.31	412.11	338.45	535.93	308.10	0.92	400.22	388.54	11.69
173	155.25	378.05	338.37	39.67	0.05	406.64	356.22	378.59	338.38	442.84	307.98	0.92	396.83	387.65	9.17
174	155.77	385.53	338.93	46.60	0.06	416.29	356.24	385.74	338.39	464.27	308.01	0.92	398.23	387.65	10.57
175	156.63	399.85	338.68	60.31	0.05	435.59	356.27	400.06	338.42	507.12	308.07	0.91	400.22	388.54	11.69
176	156.97	404.12	388.52	15.60	0.05	474.20	356.34	428.68	338.47	592.80	308.15	0.92	404.12	388.54	15.58
177	152.56	338.49	315.04	23.45	0.06	370.87	356.15	349.58	338.31	343.32	307.78	0.92	341.88	311.95	29.93
178	153.59	350.57	320.59	29.98	0.06	380.53	356.17	356.74	338.33	364.75	307.84	0.94	351.82	318.75	33.07
179	154.97	368.90	332.47	36.42	0.06	399.83	356.21	371.06	338.37	407.58	307.92	0.95	376.76	346.55	30.21
180	156.74	395.16	367.54	27.62	0.05	438.43	356.28	399.69	338.43	493.28	308.05	0.94	397.30	388.54	8.76
181	153.09	346.30	319.51	26.79	0.05	375.98	356.16	353.73	338.32	357.54	307.82	0.94	346.89	317.15	29.75
182	153.85	357.42	324.51	32.92	0.05	385.64	356.18	360.89	338.34	378.96	307.86	0.92	358.05	325.53	32.51
183	155.20	374.05	335.07	38.98	0.06	404.94	356.22	375.20	338.38	421.81	307.95	0.94	384.92	364.50	20.43
184	156.75	397.63	376.53	21.10	0.06	443.54	356.29	403.83	338.43	507.50	308.07	0.94	398.27	388.54	9.74
185	154.08	359.10	324.27	34.83	0.06	386.21	356.18	362.01	338.34	385.97	307.88	0.94	361.31	323.72	37.59
186	154.68	368.25	330.13	38.11	0.06	395.86	356.20	369.17	338.36	407.39	307.92	0.94	383.13	351.24	31.88
187	155.77	383.30	337.34	45.97	0.05	415.16	356.24	383.48	338.39	450.24	307.99	0.94	396.32	388.53	7.80
188	156.99	400.22	388.52	11.70	0.05	453.76	356.31	412.11	338.45	535.93	308.10	0.94	400.22	388.54	11.69
189	155.22	378.05	338.37	39.67	0.06	406.64	356.22	378.59	338.38	442.84	307.98	0.94	396.83	387.65	9.17
190	155.76	385.53	338.93	46.60	0.06	416.29	356.24	385.74	338.39	464.27	308.01	0.94	398.23	387.65	10.57
191	156.88	399.85	338.98	60.87	0.06	435.59	356.27	400.06	338.42	507.12	308.07	0.94	400.22	388.54	11.69
192	157.04	404.12	388.52	15.60	0.05	474.20	356.34	428.68	338.47	592.80	308.15	0.94	404.12	388.54	15.58
193	173.20	623.86	507.74	30.66	0.06	727.98	713.24	688.87	677.58	651.99	616.40	0.94	647.24	449.95	36.92
194	172.11	639.63	509.02	39.48	0.06	737.64	713.26	696.03	677.59	673.42	616.44	0.94	667.45	448.25	46.78
195	173.83	663.75	506.14	49.20	0.05	756.94	713.27	710.34	677.61	716.27	616.49	0.94	688.80	465.92	52.60
196	175.75	697.08	513.91	58.51	0.05	795.53	713.31	738.97	677.64	801.96	616.58	0.95	717.10	499.37	52.39
197	172.02	635.52	510.24	36.77	0.05	733.09	713.25	693.01	677.58	666.22	616.43	0.94	661.81	446.67	46.59
198	172.80	649.43	511.57	42.05	0.06	742.74	713.26	700.16	677.59	687.64	616.46	0.94	673.96	456.86	48.25
199	175.11	670.91	507.34	51.36	0.05	762.04	713.27	714.48	677.61	730.49	616.51	0.95	696.57	470.22	56.31
200	175.88	701.32	516.14	56.74	0.06	800.65	713.32	743.11	677.65	816.17	616.59	0.94	716.83	501.94	47.66
201	173.09	653.11	511.67	41.95	0.06	743.31	713.26	701.29	677.59	694.65	616.47	0.94	675.81	473.63	48.13
202	173.52	664.39	509.21	45.82	0.06	752.96	713.26	708.45	677.60	716.08	616.49	0.94	687.70	480.73	49.34
203	174.70	682.40	511.70	51.23	0.05	772.27	713.29	722.77	677.62	758.91	616.54	0.94	704.31	493.81	53.87
204	176.33	708.87	510.70	56.92	0.06	810.87	713.33	751.40	677.66	844.61	616.62	0.94	725.44	461.82	34.12
205	173.95	676.66	512.46	48.30	0.06	763.75	713.27	717.86	677.61	751.52	616.53	0.94	694.83	500.69	49.13
206	174.59	685.11	511.04	50.84	0.05	773.41	713.29	725.02	677.62	772.94	616.55	0.94	701.58	504.95	49.24
207	175.58	699.47	508.40	55.46	0.06	792.71	713.31	739.34	677.64	815.79	616.59	0.95	713.34	499.44	41.92
208	176.83	720.79	511.67	56.13	0.05	831.31	713.35	767.96	677.67	901.48	616.66	0.94	739.70	419.20	20.68
209	171.50	639.84	540.72	31.39	0.05	727.98	713.24	688.87	677.58	651.99	616.40	0.94	652.29	527.61	36.87
210	172.64	656.30	545.74	41.92	0.05	737.64	713.26	696.03	677.59	673.42	616.44	0.94	670.24	541.02	39.32
211	173.91	681.42	566.98	52.70	0.06	756.94	713.27	710.34	677.61	716.27	616.49	0.95	691.00	563.05	50.30
212	176.05	716.46	595.55	62.84	0.05	795.53	713.31	738.97	677.64	801.96	616.58	0.95	727.18	575.05	69.54
213	172.28	651.79	547.50	38.44	0.06	733.09	713.25	693.01	677.58	666.22	616.43	0.94	664.39	543.29	41.87
214	173.11	666.59	557.22	44.37	0.05	742.74	713.26	700.16	677.59	687.64	616.46	0.95	676.77	552.17	41.6

	optimal				Heuristic 1			Heuristic 2		Heuristic 3		Decomposition Heuristic			
	time (sec)	total cost	energy cost	switch. cost	time (sec)	total cost	energy cost	total cost	energy cost	total cost	energy cost	time (sec)	total cost	energy cost	switch. cost
217	173.13	670.51	564.15	44.57	0.05	743.31	713.26	701.29	677.59	694.65	616.47	0.95	680.65	552.21	44.65
218	173.72	682.35	571.79	49.45	0.06	752.96	713.26	708.45	677.60	716.08	616.49	0.95	692.46	562.16	51.44
219	175.08	701.44	591.76	56.57	0.05	772.27	713.29	722.77	677.62	758.91	616.54	0.95	709.77	586.25	58.23
220	176.86	730.17	605.64	63.60	0.06	810.87	713.33	751.40	677.66	844.61	616.62	0.97	738.72	573.77	70.50
221	174.45	695.72	594.24	50.85	0.06	763.75	713.27	717.86	677.61	751.52	616.53	0.95	703.69	569.24	56.57
222	174.69	704.80	593.81	57.06	0.06	773.41	713.29	725.02	677.62	772.94	616.55	0.95	712.77	585.80	58.46
223	175.72	720.45	600.22	60.36	0.05	792.71	713.31	739.34	677.64	815.79	616.59	0.97	727.04	581.18	63.44
224	177.28	744.50	600.69	70.93	0.05	831.31	713.35	767.96	677.67	901.48	616.66	0.95	755.69	573.22	84.36
225	171.63	650.71	610.22	31.13	0.06	727.98	713.24	688.87	677.58	651.99	616.40	0.95	651.98	616.77	35.21
226	172.72	667.39	613.18	41.73	0.06	737.64	713.26	696.03	677.59	673.42	616.44	0.95	669.65	630.97	38.68
227	174.23	692.78	624.93	53.72	0.05	756.94	713.27	710.34	677.61	716.27	616.49	0.97	694.93	632.96	55.76
228	176.16	731.07	624.60	69.08	0.05	795.53	713.31	738.97	677.64	801.96	616.58	0.97	735.55	641.97	80.50
229	172.44	661.58	616.99	36.57	0.06	733.09	713.25	693.01	677.58	666.22	616.43	0.97	664.31	619.40	44.91
230	173.16	677.21	619.79	46.15	0.05	742.74	713.26	700.16	677.59	687.64	616.46	0.95	678.75	630.98	47.77
231	174.56	700.79	628.31	59.56	0.05	762.04	713.27	714.48	677.61	730.49	616.51	0.97	703.43	631.30	64.20
232	176.23	736.26	629.23	67.03	0.06	800.65	713.32	743.11	677.65	816.17	616.59	0.98	742.59	641.50	87.00
233	173.36	680.11	621.32	49.29	0.05	743.31	713.26	701.29	677.59	694.65	616.47	0.97	681.18	631.28	49.63
234	173.81	692.74	629.29	52.74	0.06	752.96	713.26	708.45	677.60	716.08	616.49	0.97	694.38	635.91	58.19
235	174.99	714.41	636.33	65.05	0.06	772.27	713.29	722.77	677.62	758.91	616.54	0.98	716.98	644.09	65.15
236	176.84	745.90	632.54	70.11	0.06	810.87	713.33	751.40	677.66	844.61	616.62	0.97	751.98	649.43	89.22
237	174.30	707.31	635.22	61.53	0.05	763.75	713.27	717.86	677.61	751.52	616.53	0.97	710.64	641.14	69.50
238	174.95	718.05	639.08	68.24	0.06	773.41	713.29	725.02	677.62	772.94	616.55	0.97	720.81	652.19	68.42
239	175.95	735.51	627.05	69.13	0.06	792.71	713.31	739.34	677.64	815.79	616.59	0.97	737.73	651.81	79.48
240	177.33	762.48	635.42	81.57	0.06	831.31	713.35	767.96	677.67	901.48	616.66	0.99	782.89	717.35	54.43
241	170.27	651.18	619.82	31.36	0.05	727.98	713.24	688.87	677.58	651.99	616.40	0.95	651.99	616.40	35.59
242	171.49	667.91	620.29	40.62	0.06	737.64	713.26	696.03	677.59	673.42	616.44	0.95	674.55	605.40	53.24
243	172.14	692.66	630.87	52.50	0.06	756.94	713.27	710.34	677.61	716.27	616.49	0.97	702.83	616.80	63.82
244	173.89	729.97	639.55	69.57	0.06	795.53	713.31	738.97	677.64	801.96	616.58	0.97	747.71	665.54	56.22
245	170.84	662.08	625.25	36.81	0.06	733.09	713.25	693.01	677.58	666.22	616.43	0.95	666.22	616.43	49.79
246	171.55	677.67	625.07	44.77	0.06	742.74	713.26	700.16	677.59	687.64	616.46	0.95	686.47	612.97	55.75
247	172.45	700.17	630.04	56.90	0.06	762.04	713.27	714.48	677.61	730.49	616.51	0.97	708.58	619.48	66.23
248	174.19	735.36	641.24	72.47	0.06	800.65	713.32	743.11	677.65	816.17	616.59	0.95	751.97	674.12	57.90
249	171.73	680.65	630.29	48.93	0.05	743.31	713.26	701.29	677.59	694.65	616.47	0.95	684.77	627.61	56.72
250	172.17	693.07	632.41	50.43	0.06	752.96	713.26	708.45	677.60	716.08	616.49	0.95	696.16	625.51	54.84
251	173.14	713.52	635.31	60.67	0.05	772.27	713.29	722.77	677.62	758.91	616.54	0.95	716.33	639.06	62.72
252	174.63	744.96	648.14	74.80	0.05	810.87	713.33	751.40	677.66	844.61	616.62	0.97	765.95	691.62	53.48
253	172.69	707.27	641.71	58.96	0.05	763.75	713.27	717.86	677.61	751.52	616.53	0.95	712.52	633.30	72.38
254	173.14	717.37	636.45	62.92	0.05	773.41	713.29	725.02	677.62	772.94	616.55	0.95	723.52	638.36	70.52
255	174.06	734.02	642.05	69.75	0.06	792.71	713.31	739.34	677.64	815.79	616.59	0.95	739.24	641.27	81.34
256	175.27	762.14	639.23	81.09	0.05	831.31	713.35	767.96	677.67	901.48	616.66	0.97	792.73	772.04	15.46
257	170.27	651.18	619.82	31.36	0.05	727.98	713.24	688.87	677.58	651.99	616.40	0.95	651.99	616.40	35.59
258	171.13	668.16	625.79	42.37	0.05	737.64	713.26	696.03	677.59	673.42	616.44	0.97	673.42	616.44	56.98
259	172.28	694.12	639.84	54.28	0.05	756.94	713.27	710.34	677.61	716.27	616.49	0.97	698.30	635.17	63.13
260	174.00	732.86	651.79	72.46	0.05	795.53	713.31	738.97	677.64	801.96	616.58	0.97	748.94	678.99	54.20
261	170.83	662.08	625.26	36.82	0.06	733.09	713.25	693.01	677.58	666.22	616.43	0.95	666.22	616.43	49.79
262	171.41	677.98	631.01	46.97	0.05	742.74	713.26	700.16	677.59	687.64	616.46	0.97	684.77	624.82	59.94
263	172.94	702.11	641.83	59.77	0.05	762.04	713.27	714.48	677.61	730.49	616.51	0.97	704.24	632.51	64.17
264	174.11	738.13	658.85	70.72	0.06	800.65	713.32	743.11	677.65	816.17	616.59	0.97	755.25	677.41	60.11
265	171.77	680.73	630.99	49.73	0.06	743.31	713.26	701.29	677.59	694.65	616.47	0.97	684.64	627.80	56.84
266	172.06	693.60	639.93	53.65	0.06	752.96	713.26	708.45	677.60	716.08	616.49	0.97	694.79	635.21	59.58
267	173.03	715.68	647.23	64.87	0.06	772.27	713.29	722.77	677.62	758.91	616.54	0.97	717.71	643.68	64.84
268	174.70	748.14	659.56	78.89	0.05	810.87	713.33	751.40	677.66	844.61	616.62	0.97	771.07	715.02	41.08
269	172.70	708.21	645.31	61.67	0.06	763.75	713.27	717.86	677.61	751.52	616.53	0.97	712.81	636.18	76.36
270	173.38	718.93	648.59	68.28	0.06	773.41	713.29	725.02	677.62	772.94	616.55	0.97	720.78	652.28	68.45
271	173.92	736.75	653.74	73.23	0.06	792.71	713.31	739.34	677.64	815.79	616.59	0.98	739.13	647.71	76.64
272	175.41	766.01	664.10	87.46	0.05	831.31	713.35	767.96	677.67	901.48	616.66	0.97	793.60	772.13	15.71
273	170.30	651.18	619.82	31.36	0.06	727.98	713.24	688.87	677.58	651.99	616.40	0.97	651.99	616.40	35.59
274	171.13	668.16	625.79	42.37	0.06	737.64	713.26	696.03	677.59	673.42	616.44	0.97	673.42	616.44	56.98
275	172.30	694.12	639.84	54.29	0.05	756.94	713.27	710.34	677.61	716.27	616.49	0.98	698.30	635.17	63.13
276	173.98	733.39	658.18	75.21	0.05	795.53	713.31	738.97	677.64	801.96	616.58	0.98	748.37	693.61	54.77
277	170.84	662.08	625.26	36.82	0.05	733.09	713.25	693.01	677.58	666.22	616.43	0.97	666.22	616.43	49.79
278	171.44	677.98	631.01	46.97	0.05	742.74	713.26	700.16	677.59	687.64	616.46	0.98	684.77	624.82	59.94
279	172.61	702.13	642.09	60.04	0.06	762.04	713.27	714.48	677.61	730.49	616.51	0.99	703.65	639.23	64.41
280	173.98	738.77	665.83	72.94	0.06	800.65	713.32	743.11	677.65	816.17	616.59	0.98	754.36	693.62	60.75
281	171.77	680.73	631.00	49.73	0.05	743.31	713.26	701.29	677.59	694.65	616.47	0.97	684.64	627.80	56.84
282	172.08	693.60	639.94	53.66	0.06	752.96	713.26	708.45	677.60	716.08	616.49	0.98	694.79	635.21	59.58
283	173.08	715.85	649.93	65.92	0.06	772.27	713.29	722.77	677.62	758.91	616.54	0.99	717.09	651.98	65.11
284	174.83	749.09	670.86	77.84	0.05	810.87	713.33	751.40	677.66	844.61	616.62	0.98	770.70	729.48	41.14
285	172.72	708.25	645.93	62.33	0.06	763.75	713.27	717.86	677.61	751.52	616.53	0.99	712.74	636.30	76.44
286	173.33	719.19	649.44	69.75	0.05	773.41	713.29	725.02	677.62	772.94	616.55	0.98	720.76	652	

	total cost	energy cost	switching cost	% cost saving	% energy saving
1	145.99	132.67	13.32	27.73	33.67
2	153.03	140.91	12.12	24.62	29.55
3	183.54	179.06	4.48	10.47	10.47
4	192.85	184.54	8.30	7.73	7.73
5	170.71	156.86	13.85	15.91	21.57
6	199.93	196.01	3.92	1.99	1.99
7	190.08	184.54	5.54	7.73	7.73
8	191.83	182.69	9.14	8.65	8.65
9	196.91	192.11	4.80	3.94	3.94
10	201.89	196.01	5.88	1.99	1.99
11	199.80	192.11	7.68	3.94	3.94
12	195.61	184.54	11.07	7.73	7.73
13	192.80	183.62	9.18	7.75	8.19
14	201.72	192.11	9.61	3.94	3.94
15	189.80	179.06	10.74	10.47	10.47
16	203.35	188.29	15.06	5.86	5.86
17	159.84	147.09	12.75	20.87	26.46
18	178.97	164.37	14.59	11.84	17.81
19	187.26	182.69	4.57	8.65	8.65
20	198.75	190.19	8.56	4.91	4.91
21	168.53	148.97	19.56	16.98	25.51
22	190.13	186.40	3.73	6.80	6.80
23	186.29	180.87	5.43	9.57	9.57
24	193.77	184.54	9.23	7.73	7.73
25	189.13	183.62	5.51	7.74	8.19
26	199.87	194.05	5.82	2.97	2.97
27	191.92	184.54	7.38	7.73	7.73
28	201.60	190.19	11.41	4.91	4.91
29	208.94	198.99	9.95	0.03	0.50
30	199.70	190.19	9.51	4.91	4.91
31	195.61	184.54	11.07	7.73	7.73
32	199.30	184.54	14.76	7.73	7.73
33	156.13	145.73	10.40	22.71	27.13
34	149.50	129.74	19.76	26.35	35.13
35	204.99	199.99	5.00	0.00	0.00
36	202.78	194.05	8.73	2.97	2.97
37	179.55	165.33	14.22	11.55	17.34
38	190.13	186.40	3.73	6.80	6.80
39	188.18	182.69	5.48	8.65	8.65
40	195.73	186.40	9.32	6.80	6.80
41	189.13	183.62	5.51	7.74	8.19
42	199.87	194.05	5.82	2.97	2.97
43	195.82	188.29	7.53	5.86	5.86
44	201.60	190.19	11.41	4.91	4.91
45	194.75	185.47	9.27	6.82	7.26
46	191.83	182.69	9.14	8.65	8.65
47	193.66	182.69	10.96	8.65	8.65
48	211.69	196.01	15.68	1.99	1.99
49	158.73	145.21	13.52	21.42	27.40
50	161.77	150.24	11.53	20.31	24.88
51	185.39	180.87	4.52	9.57	9.57
52	192.85	184.54	8.30	7.73	7.73
53	166.78	145.93	20.85	17.84	27.03
54	190.13	186.40	3.73	6.80	6.80
55	188.18	182.69	5.48	8.65	8.65
56	193.77	184.54	9.23	7.73	7.73
57	189.15	184.54	4.61	7.73	7.73
58	197.87	192.11	5.76	3.94	3.94
59	201.81	194.05	7.76	2.97	2.97
60	193.66	182.69	10.96	8.65	8.65
61	189.01	180.87	8.14	9.57	9.57
62	193.77	184.54	9.23	7.73	7.73
63	195.61	184.54	11.07	7.73	7.73
64	199.30	184.54	14.76	7.73	7.73
65	154.57	139.42	15.14	23.48	30.29
66	170.99	153.23	17.77	15.77	23.39
67	189.15	184.54	4.61	7.73	7.73
68	202.78	194.05	8.73	2.97	2.97
69	173.07	153.44	19.62	14.74	23.28
70	188.23	184.54	3.69	7.73	7.73
71	193.94	188.29	5.65	5.86	5.86
72	193.77	184.54	9.23	7.73	7.73

Table 9: Results for $N = 15$

	total cost	energy cost	switching cost	% cost saving	% energy saving
73	189.15	184.54	4.61	7.73	7.73
74	193.94	188.29	5.65	5.86	5.86
75	201.81	194.05	7.76	2.97	2.97
76	195.61	184.54	11.07	7.73	7.73
77	198.70	189.24	9.46	4.93	5.38
78	205.81	196.01	9.80	1.99	1.99
79	193.66	182.69	10.96	8.65	8.65
80	199.30	184.54	14.76	7.73	7.73
81	152.89	138.50	14.38	24.31	30.75
82	161.31	148.34	12.98	20.54	25.83
83	191.06	186.40	4.66	6.80	6.80
84	200.76	192.11	8.65	3.94	3.94
85	169.13	152.00	17.13	16.68	24.00
86	186.35	182.69	3.65	8.65	8.65
87	184.43	179.06	5.37	10.47	10.47
88	201.72	192.11	9.61	3.94	3.94
89	183.51	178.16	5.35	10.48	10.92
90	193.94	188.29	5.65	5.86	5.86
91	199.80	192.11	7.68	3.94	3.94
92	193.66	182.69	10.96	8.65	8.65
93	185.25	177.27	7.98	11.37	11.37
94	195.73	186.40	9.32	6.80	6.80
95	201.60	190.19	11.41	4.91	4.91
96	201.32	186.40	14.91	6.80	6.80
97	280.10	260.71	19.38	30.32	34.82
98	296.98	264.55	26.06	26.31	33.86
99	321.02	280.22	31.62	20.74	29.95
100	377.39	369.08	8.30	7.73	7.73
101	316.59	292.08	24.51	21.44	26.98
102	328.13	297.26	28.85	18.78	25.69
103	314.81	272.90	25.73	22.46	31.78
104	374.52	365.39	9.14	8.65	8.65
105	313.85	269.02	44.81	22.51	32.75
106	362.19	343.08	18.36	10.79	14.23
107	391.91	384.22	7.68	3.94	3.94
108	379.79	365.39	10.96	7.82	8.65
109	376.42	367.24	9.18	7.97	8.19
110	393.33	380.38	9.51	4.07	4.91
111	368.86	358.12	10.74	10.47	10.47
112	391.64	376.57	15.06	5.86	5.86
113	306.31	289.82	16.49	23.80	27.55
114	339.38	313.73	25.65	15.79	21.57
115	331.16	301.34	29.81	18.23	24.66
116	388.94	380.38	8.56	4.91	4.91
117	315.52	291.17	24.35	21.71	27.21
118	287.59	256.35	31.23	28.82	35.91
119	323.53	294.04	28.90	20.31	26.49
120	378.31	369.08	9.23	7.73	7.73
121	298.93	256.56	42.37	26.19	35.86
122	328.09	295.22	32.87	19.19	26.19
123	376.46	369.08	7.38	7.73	7.73
124	391.79	380.38	11.41	4.91	4.91
125	407.93	397.98	9.95	0.26	0.50
126	389.89	380.38	9.51	4.91	4.91
127	380.15	369.08	11.07	7.73	7.73
128	383.84	369.08	14.76	7.73	7.73
129	304.48	288.24	16.25	24.26	27.94
130	280.07	247.76	32.32	30.50	38.06
131	350.87	325.33	25.54	13.37	18.67
132	396.84	388.10	8.73	2.97	2.97
133	333.60	310.14	23.46	17.22	22.46
134	313.98	290.56	23.42	22.28	27.36
135	325.91	294.87	31.04	19.73	26.28
136	382.13	372.81	9.32	6.80	6.80
137	314.71	279.09	35.62	22.29	30.23
138	349.44	311.39	38.05	13.93	22.15
139	384.11	376.57	7.53	5.86	5.86
140	391.79	380.38	11.41	4.91	4.91
141	380.22	370.94	9.27	7.04	7.26
142	374.52	365.39	9.14	8.65	8.65
143	376.35	365.39	10.96	8.65	8.65
144	407.70	392.02	15.68	1.99	1.99

Table 10: Results for $N = 15$ (continued)

	total cost	energy cost	switching cost	% cost saving	% energy saving
145	304.34	284.09	20.25	24.29	28.98
146	304.63	279.35	25.27	24.41	30.16
147	305.73	277.81	27.92	24.51	30.55
148	377.39	369.08	8.30	7.73	7.73
149	309.30	280.39	28.91	23.25	29.90
150	345.82	330.10	15.72	14.40	17.47
151	315.51	276.00	34.82	22.29	31.00
152	378.31	369.08	9.23	7.73	7.73
153	310.47	275.79	34.67	23.34	31.05
154	335.20	287.34	47.87	17.44	28.17
155	395.86	388.10	7.76	2.97	2.97
156	376.35	365.39	10.96	8.65	8.65
157	369.88	361.74	8.14	9.57	9.57
158	378.31	369.08	9.23	7.73	7.73
159	380.15	369.08	11.07	7.73	7.73
160	383.84	369.08	14.76	7.73	7.73
161	297.10	274.63	22.48	26.09	31.34
162	321.73	294.34	27.38	20.17	26.41
163	328.37	299.05	29.32	18.92	25.24
164	396.84	388.10	8.73	2.97	2.97
165	322.06	295.34	26.72	20.08	26.16
166	274.04	246.08	27.96	32.17	38.48
167	326.86	299.52	27.34	19.49	25.12
168	378.31	369.08	9.23	7.73	7.73
169	299.70	259.49	40.21	26.00	35.13
170	326.23	287.02	39.21	19.65	28.24
171	395.86	388.10	7.76	2.97	2.97
172	380.15	369.08	11.07	7.73	7.73
173	387.94	378.48	9.46	5.15	5.38
174	401.82	392.02	9.80	1.99	1.99
175	376.35	365.39	10.96	8.65	8.65
176	383.84	369.08	14.76	7.73	7.73
177	293.87	275.49	18.38	26.90	31.13
178	306.16	283.42	22.74	24.03	29.15
179	321.54	287.00	34.54	20.61	28.25
180	392.87	384.22	8.65	3.94	3.94
181	312.28	290.61	21.67	22.51	27.35
182	315.82	287.61	28.21	21.83	28.10
183	325.40	301.84	23.56	19.85	24.54
184	393.83	384.22	9.61	3.94	3.94
185	298.80	261.02	37.78	26.22	34.74
186	356.26	330.02	26.23	12.25	17.49
187	391.91	384.22	7.68	3.94	3.94
188	376.35	365.39	10.96	8.65	8.65
189	362.51	354.54	7.98	11.37	11.37
190	382.13	372.81	9.32	6.80	6.80
191	391.79	380.38	11.41	4.91	4.91
192	387.72	372.81	14.91	6.80	6.80
193	585.77	376.17	26.89	26.96	52.98
194	577.63	406.63	25.39	28.07	49.17
195	604.62	404.27	36.16	24.89	49.47
196	660.18	467.44	52.52	18.40	41.57
197	620.09	426.87	27.07	22.78	46.64
198	632.13	376.36	33.75	21.38	52.96
199	613.88	434.94	41.64	23.84	45.63
200	610.79	487.84	46.10	24.59	39.02
201	611.74	366.36	37.58	24.01	54.21
202	683.98	509.32	40.33	15.14	36.34
203	644.39	403.23	51.55	20.25	49.60
204	661.44	536.67	42.77	18.54	32.92
205	651.03	497.45	30.53	19.53	37.82
206	675.45	528.87	40.54	16.61	33.89
207	613.20	485.96	51.05	24.48	39.26
208	665.31	519.15	70.93	18.47	35.11
209	612.24	431.11	19.48	23.66	46.11
210	663.58	501.09	24.32	17.36	37.36
211	629.08	477.20	34.71	21.85	40.35
212	646.64	471.42	54.11	20.07	41.07
213	624.24	431.27	32.55	22.26	46.09
214	569.28	414.42	36.90	29.19	48.20
215	614.78	453.84	43.92	23.72	43.27
216	624.29	474.49	49.89	22.93	40.69

Table 11: Results for $N = 15$ (continued)

	total cost	energy cost	switching cost	% cost saving	% energy saving
217	569.80	415.03	38.21	29.22	48.12
218	613.01	471.81	39.14	23.94	41.02
219	672.13	429.78	50.85	16.82	46.28
220	662.73	494.59	65.40	18.38	38.18
221	613.23	489.46	54.41	24.20	38.82
222	653.55	451.60	53.18	19.31	43.55
223	624.10	501.33	53.68	23.14	37.33
224	657.48	576.56	56.33	19.43	27.93
225	592.72	576.47	16.25	26.09	27.94
226	534.56	490.02	44.54	33.43	38.75
227	676.71	618.44	38.80	15.94	22.69
228	704.35	618.51	57.26	12.94	22.69
229	647.64	619.06	28.58	19.35	22.62
230	609.94	572.83	37.11	24.14	28.40
231	620.21	570.29	48.62	23.05	28.71
232	673.97	618.89	46.38	16.79	22.64
233	595.22	551.23	43.99	26.06	31.10
234	656.71	607.62	49.09	18.52	24.05
235	598.08	508.94	82.87	25.98	36.38
236	694.28	618.97	53.67	14.50	22.63
237	637.51	581.46	56.05	21.20	27.32
238	603.86	526.94	76.03	25.45	34.13
239	645.54	576.50	61.84	20.50	27.94
240	799.73	784.05	15.68	1.99	1.99
241	588.37	568.00	20.37	26.64	29.00
242	591.62	547.46	31.73	26.32	31.57
243	578.57	516.87	31.97	28.13	35.39
244	649.37	576.85	58.22	19.73	27.89
245	592.64	558.82	33.82	26.20	30.15
246	680.09	645.20	24.79	15.41	19.35
247	588.53	520.92	47.99	26.98	34.89
248	648.32	590.77	38.69	19.96	26.15
249	587.54	543.86	43.61	27.01	32.02
250	624.97	530.99	55.93	22.46	33.63
251	643.14	572.06	53.14	20.40	28.49
252	657.35	597.36	51.22	19.05	25.33
253	603.27	535.87	62.44	25.43	33.02
254	582.09	507.76	62.91	28.14	36.53
255	657.45	588.67	57.58	19.03	26.42
256	750.56	730.78	14.62	8.02	8.65
257	571.73	549.26	22.48	28.71	31.34
258	622.46	582.71	39.74	22.48	27.16
259	650.02	570.93	34.49	19.25	28.63
260	689.65	580.58	58.70	14.75	27.43
261	620.22	588.80	31.42	22.76	26.40
262	525.94	483.72	42.22	34.58	39.53
263	612.09	562.24	41.67	24.06	29.72
264	695.48	640.44	42.29	14.14	19.94
265	570.46	516.35	54.10	29.14	35.46
266	601.44	547.44	54.00	25.38	31.57
267	631.02	556.54	53.69	21.90	30.43
268	630.58	518.09	72.51	22.34	35.24
269	628.07	571.72	55.02	22.36	28.54
270	665.72	590.77	72.54	17.81	26.15
271	637.76	569.68	64.27	21.46	28.79
272	752.93	738.16	14.76	7.73	7.73
273	569.59	550.97	18.62	28.98	31.13
274	584.23	553.48	30.75	27.24	30.82
275	602.33	559.28	43.05	25.18	30.09
276	664.80	624.42	40.35	17.83	21.95
277	607.86	579.39	28.46	24.30	27.58
278	613.32	567.88	45.45	23.72	29.02
279	612.02	564.92	47.10	24.07	29.38
280	666.18	597.75	65.37	17.76	25.28
281	567.44	520.94	46.50	29.51	34.88
282	680.28	647.82	32.46	15.60	19.02
283	658.92	599.45	59.47	18.45	25.07
284	654.04	607.16	46.83	19.45	24.10
285	597.00	541.46	55.54	26.21	32.32
286	594.88	506.20	88.68	26.56	36.72
287	637.58	572.98	64.60	21.48	28.38
288	760.53	745.62	14.91	6.80	6.80

Table 12: Results for $N = 15$ (continued)